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Reg. No. | | | | | | | |

Time : 3.00 hrs.

Quarterly Examination - 2025

PHYSICS

Max. Marks : 70

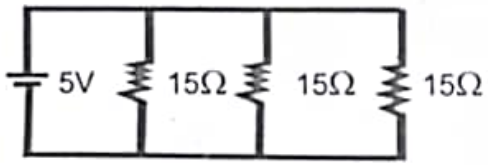
PART - A

Note: (i) Answer all the questions. **15x1=15**

(ii) Choose the most appropriate answer from the four given alternatives and write the option code with the corresponding answer.

I. Choose the best answer.

1. Which charge configuration produces a uniform electric field?
   a) Point charge b) Uniformly charged infinite line c) Uniformly charged infinite plane d) Uniformly charged spherical shell
2. A toaster operating at 240V has a resistance of 120Ω. Its power is
   a) 400W b) 2W c) 480W d) 240W
3. When the current changes from +2A to -2A in 0.05S, an emf of 8v is induced in a coil. The co-efficient of self-induction of the coil is a) 0.2H b) 0.4H c) 0.8H d) 0.1H
4. In a transformer, the number of turns in the primary and the secondary are 410 and 1230 respectively. If the current in primary is 6A, then that in the secondary coil is a) 2A b) 18A c) 12A d) 1A
5. A circular coil of radius 5cm and 50 turns carries a current of 3 ampere. The magnetic dipole moment of the coil is nearly a) 1.0Am^2 b) 1.2Am^2 c) 0.5Am^2 d) 0.8Am^2
6. Calculate the number of electrons in a charge of 2 coulomb.
   a) 0.25×10^{19} electrons b) 12.5×10^{19} electrons c) 1.25×10^{19} electrons d) 0.125×10^{19} electrons
7. Which of the following is an electromagnetic wave?
   a) α - rays b) β - rays c) γ - rays d) all of them
8. The dimension of $\frac{1}{\mu_o\varepsilon_o}$ is a) $[LT^{-1}]$ b) $[L^2T^{-2}]$ c) $[L^{-1}T]$ d) $[L^{-2}T^2]$
9. The total electric flux for the closed surface which is kept inside water
   a) $\frac{80q}{\varepsilon_o}$ b) $\frac{q}{40\varepsilon_o}$ c) $\frac{q}{80\varepsilon_o}$ d) $\frac{q}{160\varepsilon_o}$
10. What is the current drawn out from the battery?
    a) 1A b) 2A c) 3A d) 4A



$\vec{B} = 0.2\hat{i}\ T$

11. The potential energy of magnetic dipole whose dipole moment is $\vec{Pm} = (-0.5\hat{i} + 0.4\hat{j})$ Am^2 kept in uniform magnetic field .... a) -0.1J b) -0.8J c) 0.1J d) 0.8J
12. The phase difference between V_L and V_c in series RLC circuit is a) 2π b) $\frac{\pi}{2}$ c) $\frac{2\pi}{3}$ d) π
13. If the amplitude of the magnetic field is 3×10^{-6}T, then amplitude of the electric field for a electromagnetic wave is a) 100vm^{-1} b) 300vm^{-1} c) 600vm^{-1} d) 900vm^{-1}
14. Two wires of A and B with circular cross section are made up of the same material with equal lengths. Suppose $R_A=3R_B$, then what is the ratio of radius of wire A to that of B? a) $\sqrt{3}$ b) 3 c) $\frac{1}{\sqrt{3}}$ d) $\frac{1}{3}$
15. One weber is equal to a) 10^2 Maxwell b) 10^6 Maxwell c) 10^4 Maxwell d) 10^8 Maxwell

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PART - II

II. Answer any six of the following questions. Q.No.24 is compulsory. 6x2=12

16. Define electric field. Write it's unit.
17. What is electric power and electric energy?
18. Find the heat energy produced in a resistance of 10Ω, when 5A current flows through it for 5 minutes.
19. What is an electric dipole moment? Write it's unit.
20. Give the uses of Infra red radiation.
21. State Fleming's left hand rule.
22. Why are electromagnetic waves non-mechanical?
23. What are the advantages of Alternating Current System over Direct Current System?
24. A straight metal wire crosses a magnetic field of flux 4 mwb in a time 0.4S. Find the magnitude of the emf induced in the wire.

PART - III

III. Answer any six of the following questions. Q.No.33 is compulsory. 6x3=18

25. What are the differences between Coulomb force and gravitational force?
26. How will you induce an emf by changing the area enclosed by the coil?
27. Why is the path of a charged particle not a circle when its velocity is not perpendicular to the magnetic field?
28. State the applications of Seebeek effect.
29. Dielectric strength of air is $3x10^{6}vm^{-1}$. Suppose the radius of a hollow sphere in the Van de Graff generator is R=0.5m, calculate the maximum potential difference created by this Van de Graff generator.
30. How will you define Q-factor?
31. What are Fraunhofer lines? How are they useful in the identification of elements present in the Sun?
32. How can the current sensitivity of the galvanometer increased?
33. The resistance of a nichrome wire at 20°C is 10Ω. If its temperature coefficient of resistivity of nichrome is 0.004/°C, find the resistance of the wire at boiling point of water. Comment on the result.

PART - IV

IV. Answer all the questions. 5x5=25

34. Calculate the electric field due to a dipole on its axial line. **(OR)**
    Derive the expression for the force on a current carrying conductor in a magnetic field. Write its special cases also.
35. Find out the phase relationship between Voltage and Current in a pure inductive circuit. **(OR)**
    Explain the determination of the internal resistance of a cell using voltmeter.
36. Explain the principle, construction and working of transformer. **(OR)**
    Explain the equivalent resistance of a series and parallel resistor network.
37. What is meant by spectrum. What is emission spectrum? Explain the types of emission spectrum. **(OR)**
    Define Capacitance. Obtain the expression for capacitance for a parallel plate capacitor.
38. Using Biot-Savart Law Deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current. **(OR)**
    (i) Write down the properties of electromagnetic waves.(any six properties)
    (ii) If the relative permeability and relative permittivity of a medium are 1.0 and 2.25 respectively, find the speed of the electromagnetic wave in the medium.

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